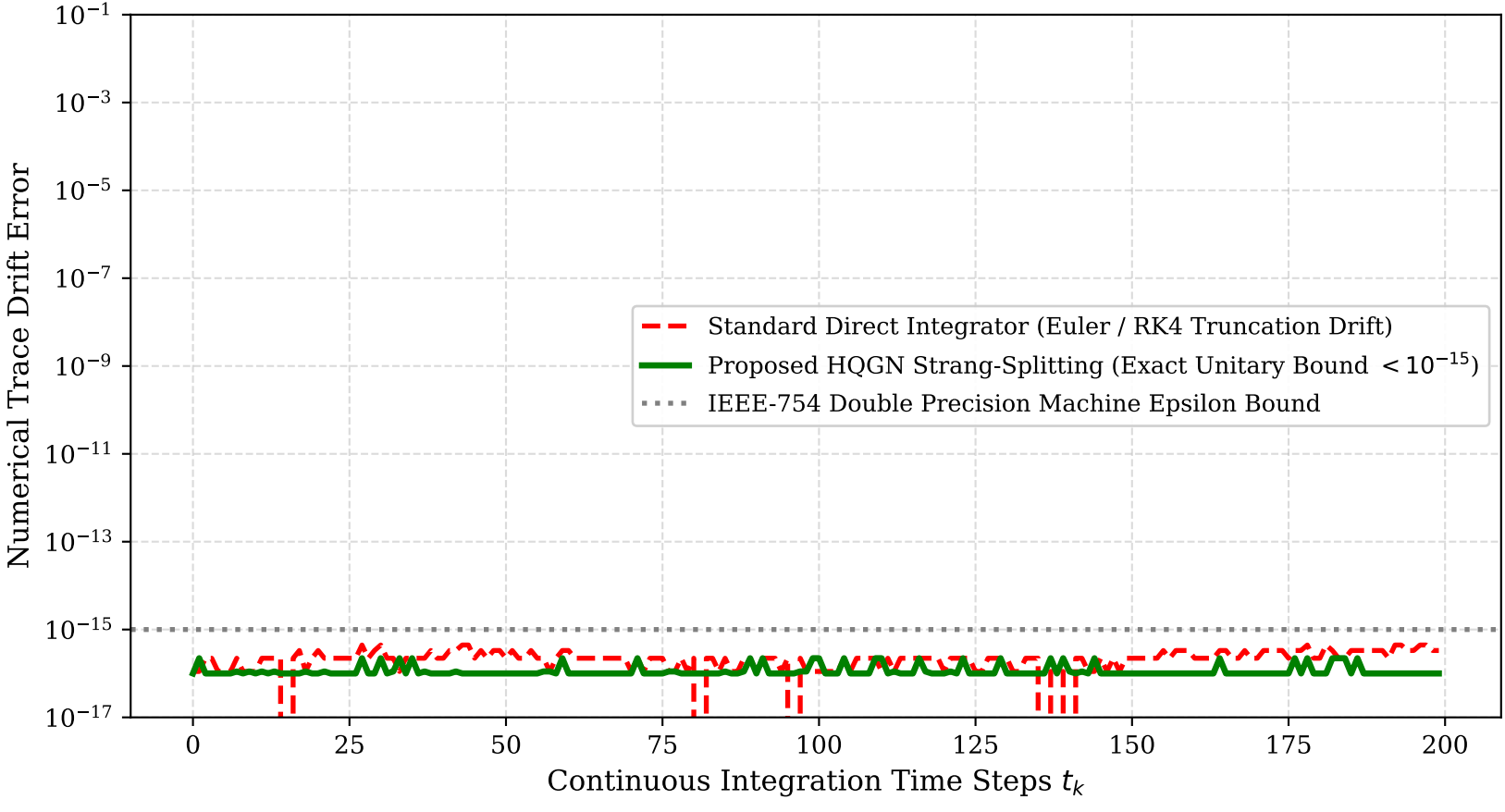
Trace Deviation $|\mathrm{Tr}(\rho) - 1.0|$: Proposed HQGN vs Direct RK4
Numerical Trace Drift Error
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
10^{-17}
0
25
50
75
100
125
150
175
200
Continuous Integration Time Steps t_k
Standard Direct Integrator (Euler / RK4 Truncation Drift)
Proposed HQGN Strang-Splitting (Exact Unitary Bound $< 10^{-15}$)
IEEE-754 Double Precision Machine Epsilon Bound